



KiCad Custom Symbol & Footprint Creation

When the vendor part isn't in your library

Symbol Editor + Footprint Editor — the deep skills

Audience: KiCad Intermediate Users

Symbol Editor • Footprint Editor • IPC-7351 • 3D models • Verification



When to Make Your Own

- ▶ Manufacturer doesn't provide a KiCad library
- ▶ Vendor library has bugs (wrong pin assignment, wrong footprint)
- ▶ Need custom symbol grouping (e.g., split a large MCU into multiple symbols)
- ▶ Pre-existing component sources (SnapEDA, Component Search Engine) lack the part
- ▶ Want full control over symbol look-and-feel (corporate library)
- ▶ Modifying an existing footprint (e.g., add thermal pad pattern)

Library Architecture

- ▶ KiCad library = collection of symbols (.kicad_sym files) and footprints (.kicad_mod / .pretty)
- ▶ Symbol library: .kicad_sym file contains multiple symbols
- ▶ Footprint library: .pretty directory containing individual .kicad_mod files
- ▶ Personal vs Global library: Personal in project folder, Global in user/system path
- ▶ Vendor libraries: ST, NXP, Espressif provide KiCad-format libs (sometimes)
- ▶ Open-source aggregators: SnapEDA, Ultra Librarian, Component Search Engine
- ▶ Best practice: own personal library per project (or organisation)

Symbol Editor Workflow

- ▶ Open KiCad → Symbol Editor
- ▶ File → New Library OR Open existing personal lib
- ▶ File → New Symbol → name + ref designator prefix (e.g., U for IC)
- ▶ Add pins via Place → Pin (P)
- ▶ Pin properties: number, name, electrical type (input/output/power/passive)
- ▶ Pin position: align to schematic grid (default 50 mil)
- ▶ Draw the body: lines, rectangles, polygons
- ▶ Save symbol → File → Save All
- ▶ Test: place in schematic, verify pin numbers match datasheet

Footprint Editor Workflow

- ▶ Open KiCad → Footprint Editor
- ▶ File → New Library OR open existing personal lib
- ▶ File → New Footprint → name (match symbol naming)
- ▶ Add pads via Place → Pad (P) — Through-hole, SMD, etc.
- ▶ Pad properties: number, shape (round/rect/oval), size, drill (THT)
- ▶ Reference position from IPC-7351 landpattern guidance
- ▶ Draw silkscreen + courtyard + body outline
- ▶ Add fabrication / assembly layer markings
- ▶ Add 3D model link (STEP / WRL file)
- ▶ Save footprint → test by importing into PCB editor

IPC-7351 Landpattern Compliance

- ▶ IPC-7351 specifies pad geometry for every standard package family
- ▶ Three density classes: Level A (most material), B (nominal), C (least)
- ▶ Use Level B for general products; Level A for ease of soldering; Level C for dense designs
- ▶ Calculator: enter package dims (per datasheet) → get pad dimensions
- ▶ Free tools: PCBLibraries online calculator, Saturn PCB Toolkit
- ▶ Most KiCad built-in footprints follow IPC-7351 Level B by default
- ▶ Custom footprints: cite which level you used in the footprint description
- ▶ Verify against vendor's datasheet 'recommended landpattern' section

3D Models

- ▶ KiCad supports STEP and WRL 3D models
- ▶ STEP (.stp/.step): industry-standard CAD format — best for mechanical engineers
- ▶ WRL (.wrl): VRML format — used by KiCad's built-in 3D viewer
- ▶ Add via Footprint Editor: Edit Footprint Properties → 3D Models tab → Add
- ▶ Path: \${KICAD8_3DMODEL_DIR}/ for built-in models
- ▶ Vendor STEP models: Component Search Engine, SnapEDA, manufacturer websites
- ▶ FreeCAD plug-in: convert your own 3D models to KiCad format
- ▶ Export full board STEP for mechanical enclosure design (Fusion 360, SolidWorks)

Symbol + Footprint Best Practices

- ▶ Naming: include package + variant in name (e.g., 'STM32F407-LQFP100')
- ▶ Pin names match datasheet exactly
- ▶ Pin numbers match physical chip pin (NOT logical order)
- ▶ Use proper electrical types for ERC
- ▶ Add pin function as comment/description
- ▶ Symbol body large enough for all pins + labels
- ▶ Footprint courtyard: actual physical body + 0.25 mm clearance
- ▶ Silkscreen body outline matches IC body
- ▶ Pin 1 marker: dot AND chamfered corner of silk body
- ▶ Include 3D model whenever possible
- ▶ Verify against datasheet — pad sizes, hole sizes, pitches
- ▶ Document source: datasheet rev + URL in description field

Common Mistakes

- ▶ Pin numbers mismatched — chip pin 1 doesn't go to symbol pin 1
- ▶ Wrong package: TQFP-32 footprint assigned to SOIC-32 symbol
- ▶ IPC-7351 landpattern compute error — pad too small for the device
- ▶ Thermal pad missing on QFN/DFN — IC overheats
- ▶ No courtyard — DRC misses placement collisions
- ▶ Pin types misassigned — ERC misses real errors
- ▶ Reused library across projects without checking pin compatibility
- ▶ Vendor library bug (e.g., MPN encoded wrong) — board doesn't work

Library Management

- ▶ Personal library per project: in project folder
- ▶ Corporate library: in shared Git repo (Git submodule for projects)
- ▶ KiCad built-in library: don't modify; copy parts to personal lib if customising
- ▶ Backup: symbol + footprint files are text — commit to Git
- ▶ Naming convention: '<manufacturer>_<MPN>_<package>' for unique identification
- ▶ Sub-libraries by category: Resistors, Capacitors, MCU, Sensors, Connectors
- ▶ Migration: KiCad upgrade may require library format conversion

Tools That Generate Libraries

SnapEDA

Web service
5M+ KiCad parts.
Free (limited).

Ultra Librarian

Web + desktop
Similar to SnapEDA.
Free.

Component Search Engine

Vendor-supplied parts.
Free.

KiCad Lib Builder

Open-source tool
for systematic creation.

FreeCAD plugin

Convert 3D models
to KiCad format.

ChatGPT/Claude

Generate symbol/
footprint files programmatically.
Verify everything!

Verification Workflow

- ▶ Place new symbol in a test schematic
- ▶ Verify pin numbers + types match datasheet
- ▶ Assign your new footprint
- ▶ Open in PCB editor, check footprint matches body of device
- ▶ Print 1:1 scale paper copy → put real component on it → measure
- ▶ Render in 3D viewer → check component fits
- ▶ Add to your project — small test board → fab + assemble → verify

Key Takeaways

Key takeaways from this lecture

Symbol = function

Pin numbers + types matter for ERC.

Footprint = physical

IPC-7351 compliance is the standard.

3D model = optional but valuable

For mechanical fit checks.

Personal library per project

Git-tracked, reproducible.

Verify against datasheet

Wrong footprint = scrap board.

Use generators carefully

SnapEDA etc. — still verify.

Questions & Discussion

Ask anything — concepts, projects, sourcing, career.

References & Further Learning

- ▶ KiCad documentation — Symbol + Footprint Editor
- ▶ IPC-7351 landpattern standard
- ▶ SnapEDA, Ultra Librarian, Component Search Engine
- ▶ Phil's Lab YouTube — KiCad symbol creation
- ▶ Robert Feranec — library management tutorials
- ▶ KiCad Forum + Reddit r/KiCad
- ▶ Vendor application notes on landpattern design
- ▶ PCB Libraries Inc. — premium landpattern tooling

Thank you • Build something this week.